

Supplementary Information

Not all generalisation failures can be bought back: four boundaries in affective audio modelling

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Sections S1–S6 are admission criteria: checks that had to pass before a main-text claim was allowed to be made, each reported in the same four parts — what was checked, the decision rule fixed before the check, the outcome, and the claim the outcome licenses. Sections S7–S9 then cover the representation-family extraction, the withdrawn analyses, and the Extended Data figures.

Table S1: The six admission criteria and the main-text claims they license. A criterion that failed removes a claim rather than weakening it; two of the six did fail, and the corresponding analyses are absent from the main text rather than reported as null results.

Section	What had to pass	What the outcome licenses
S1	The documented mapping from stimulus trigger code to audio file in ds002721 must reproduce, or be recoverable against a permutation null.	Failed. No acoustic-to-physiological analysis is reported in that corpus. Its trigger codes are used only as stimulus identifiers, for which their internal consistency suffices.
S2	Both pre-declared electrodermal positive controls must pass, applied at the level at which the conclusion would be drawn.	Failed. The corpus and its conclusion are withdrawn. The physiological negatives that remain in the main text are the ones whose instrument was shown to work.
S3	Loudness must not act as a corpus identifier before descriptors are computed; and the physiological null must not be an artefact of having removed loudness.	Boundary 1, the domain wall (Fig. 1 and Table 1); Boundary 3, the physiological ceiling.
S4	The reliability of each target must be estimated, with a stated estimator, before any percentage of it is quoted.	The denominator of every survival fraction at all four boundaries, including the 84% within corpus and the 31% bound on the physiological target.
S5	A within-song correlation between two autocorrelated series must exceed a circular-shift null before it is called a coupling.	That all four boundaries are stated at the level of the excerpt, and that moment-to-moment modulation is a separate and much smaller quantity.
S6	A cross-model attribution consensus must exceed its label-permutation reference distribution before it is treated as a stable descriptor set.	Failed. Transfer is measured at every boundary and never inferred from which descriptors a model relies on within its own corpus (Fig. 2).

S1 Admission criterion: the trigger-to-audio mapping in ds002721

Two errors in the event annotation of this dataset were found while preparing the analyses in the main text, and both would affect anyone reusing it. We report them so that the corrections are available, and we have written to the dataset authors. The second is the admission criterion for Boundary 3: it determines which questions this corpus can be asked.

S1.1 The documented response codes are not the response

What was checked. `events.json` documents two code families as both encoding a 1–9 rating: 833–841 (“Response 01–09”) and 901–909 (“Answer 01–09”). They agree on only **7.3%** of trials, so at most one can be the rating.

Decision rule. The question can be settled without any external reference, using the internal structure of the eight emotion scales. Ratings of pleasantness and happiness must covary positively; pleasantness and fear negatively; anger and fear positively. The encoding that reproduces the expected sign on the twelve relations is the rating; if neither did, no rating analysis would be reported.

Table S2: Internal structure of the eight emotion scales under each candidate response encoding.

Expected relation	833–841	901–909
pleasant $\leftrightarrow$ happy, positive	−0.015	+ 0.503
pleasant $\leftrightarrow$ tender, positive	−0.173	+ 0.519
pleasant $\leftrightarrow$ afraid, negative	+0.075	− 0.584
angry $\leftrightarrow$ afraid, positive	+0.065	+ 0.600
happy $\leftrightarrow$ sad, negative	+0.077	− 0.299
six relations expected positive	mean −0.063, 2/6	mean +0.406, 6/6
six relations expected negative	mean +0.005, 2/6	mean −0.382, 6/6

Outcome. 901–909 carries the ratings. 833–841 shows no coherent emotional structure. All rating analyses in the main text use 901–909.

A second, smaller point: the question codes 800–807 are re-emitted when a question is redisplayed, so a rating must be matched to the nearest question code within a tolerance rather than by exact timestamp equality. Exact matching drops roughly 45% of ratings.

S1.2 The stimulus codes do not resolve to the documented audio files

What was checked. `events.json` states that stimulus codes span 301–360 and that subtracting 300 gives the filename. The observed codes span **302–657**, exactly ten per music run, strictly interleaved with the fixation and music-onset markers. Assuming the codes are stimulus + 300 yields indices into the published stimulus set that do not correspond to what participants reported:

- Participants’ eight ratings against the published mean ratings of the mapped clips: all 64 pairings, $|\rho| \leq 0.16$.

- A spectral-centroid probe against published “energy” ratings: $\rho = +0.37$.
- The same probe against participants’ “energetic” ratings: $\rho = -0.01$.

The codes are nevertheless genuine stimulus identities: participants who received the same code agree strongly in their ratings (split-half 0.80–0.87). The numbering scheme, not the codes, is what is missing.

Decision rule. We attempted to recover the mapping by matching the eight-dimensional emotion profile of each code against that of each published clip, optimising the assignment with the Hungarian algorithm. The rule fixed in advance was that the recovered assignment must beat a null formed by permuting stimulus labels within participant and regenerating the profiles; otherwise the acoustic-to-physiological link would not be examined in this corpus at all.

Outcome. The recovered assignment was not better than chance (observed mean profile correlation 0.672; null mean 0.654, 95th percentile 0.681; $p = 0.19$), and none of the recovered pairings matched the documented rule.

What it licenses. **The acoustic-to-physiological link could not be examined in this corpus, and is not examined in the main text.** What Boundary 3 reports instead is a comparison between two response channels measured on the same trials — what listeners say and what their EEG does — for which stimulus identity is sufficient and stimulus *content* is not required. Every use of these codes in the main text is as an identifier only.

A third observation, possibly related: the dataset README describes participants as hearing 40 clips, which would imply a common set. We observe 39–40 clips per participant but **307 distinct codes across the 31 participants**, with a median pairwise overlap of six clips. This is also the reason the stimulus-level reliability estimator in the main text is the one appropriate to an unbalanced design.

S2 Admission criterion: withdrawal of the electrodermal corpus

An earlier version of this work analysed an electrodermal corpus [1] and reported that single-stimulus peripheral responses carry no reproducible stimulus-level component. **That conclusion is withdrawn.** It was not withdrawn because the effect exists; it was withdrawn because we could not establish that the analysis pipeline could detect any effect at all.

S2.1 How it was found

The reported conclusion conflicts with a well-established literature on category-level skin-conductance responses to affective sounds. Prompted by that conflict, we tested the pipeline against effects known to be present.

Habituation was significant in the direction opposite to the established effect, and stimulus locking was absent. Both held after replacing our implementation with a published, validated toolbox [2].

Table S3: Group-level positive controls, under two implementations. The two share their input recordings and their event timing and differ only in the decomposition of the signal, so agreement between them bounds implementation error and is not independent corroboration of the controls themselves.

Positive control	Expected	Hand-written	NeuroKit2
Habituation across trials	$\rho < 0$	+0.113 , $p = 0.002$	+0.101 , $p = 0.015$
Post-stimulus > random window	significant	$p = 0.267$	$p = 0.357$

S2.2 Five candidate explanations, all excluded

Table S4: Candidate explanations for the control failures, and their tests.

Hypothesis	Test	Outcome
Timestamp semantics (re-sponse time, not onset)	event-related average swept over ± 16 s	no peak at any lag
Missing phasic decomposition	0.05 Hz high-pass added	reliability $-0.085 \rightarrow +0.110$; controls still fail
Ill-conditioned filter at 1 kHz	decimate to 20 Hz before filtering	8.3×10^{-7} vs 7.8×10^{-7} , no change
Implementation defect	replace with NeuroKit2	same failures
Unit scaling	rescale by 10^4 and 10^5	66 responses detected at all three scalings

The second row is itself a correction. The original extraction linearly detrended within a 6 s window, while the dominant variance in this signal is drift on the scale of minutes; trial-level responses were buried rather than absent. After adding the high-pass, stimulus-level reliability rose from -0.085 to $+0.110$ and the category-level ordering became consistent with the literature. **The strong form of the original claim — that no reproducible stimulus-level component exists — is therefore withdrawn twice over**, once for the extraction defect and once for the control failure below. What can be said is that per-stimulus reliability is low (0.110) against 0.750 for the same participants’ ratings of the same trials.

S2.3 The group-level verdict was itself tested

Decision rule. A group mean in the wrong direction is compatible with a usable subset: if some recordings failed technically, they could displace the mean while the remainder stayed valid. This dataset’s own publication documents extensive equipment problems, so the possibility was concrete. We therefore applied both controls *per participant*, with each participant’s own permutation null, and fixed the rule that the corpus would be retained only if a subset of participants passed both.

Table S5: Per-participant positive controls (94 analysable participants, 500 permutations each).

Per-participant control	Passing	Chance expectation
Habituation ($\rho < 0$ and $p < 0.05$)	8 / 94	4.7
Stimulus locking ($p < 0.05$)	4 / 94	4.7
Both	0 / 94	0.24

Outcome. Decisively, habituation was in the expected direction for only **35%** of participants, where a working measurement would put it well above 50%. The wrong direction is therefore the majority behaviour, not a minority artefact displacing a mean. No usable subset exists.

What it licenses. The corpus is excluded from every substantive analysis, and the main text reports the exclusion rather than the negative result it would otherwise have produced. This is what makes the remaining physiological results at Boundary 3 readable: the EEG corpus that carries them passes four dataset-level controls and an individual-level control before any reliability is estimated. A negative result from an instrument that has not been shown to work is not a negative result.

S2.4 What we take from it

Reliability testing and positive controls answer different questions and do not substitute for one another. Testing whether a target variable carries reproducible signal says nothing about whether the instrument works. We had performed the first and omitted the second, and the omission produced a publishable-looking negative result that we could not defend.

S3 Admission criterion: loudness normalisation

Loudness enters this work twice, in opposite roles, and both had to be settled before a main claim could be made. For the domain wall, absolute level is a corpus *identifier* that must be removed or the wall is measured against an inflated baseline. For the physiological ceiling, absolute level is the most plausible *true* driver of an autonomic response, so having removed it is the most obvious mundane explanation for a null and must be excluded.

S3.1 Loudness as a corpus fingerprint

What was checked. All audio is loudness-normalised to -23 LUFS (ITU-R BS.1770, integrated, gain scaling only, no limiting) before any descriptor is computed. The check is whether this changes cross-corpus transfer, using DEAM $\rightarrow$ PMEmo with five classifiers, and whether it changes which descriptors the models use.

Decision rule. The rule was written in terms of the loss in transfer performance that normalisation would cause, with three branches: a small loss would mean loudness carries genuine shared signal; a large loss would mean the models had been learning loudness alone and the descriptor work could not proceed; an intermediate loss would require a per-corpus decision.

Outcome. Transfer performance *rose*, which is a fourth case the rule did not cover. We record that rather than reclassify it after the fact.

The mechanism is specific and checkable: DEAM is a mixed library and PMEmo consists of pop choruses mastered at high level, so the two corpora differ in loudness distribution before they differ in anything else. A model trained on the first learns that loud means aroused and

Table S6: Loudness normalisation, DEAM $\rightarrow$ PMEmo. The first three rows are out-of-corpus quantities and all improve; the fourth is a within-corpus sanity check and falls slightly. The last row gives the mean rank, across models, of the strongest absolute-level descriptor among the 122.

	Raw	−23 LUFS	Δ
Cross-corpus AUC (random forest)	0.9180	0.9304	+0.0124
Cross-corpus recall (random forest)	0.4138	0.5172	+0.1034
Test excerpts inside the applicability domain	78.0%	86.7%	+8.7 pp
Within-corpus cross-validated AUC	0.9378	0.9282	−0.0096
Mean attribution rank of <code>rms_p90</code> (of 122)	39.0	93.0	+54.0

mis-calibrates on the second. Normalisation removes that layer, which is why an out-of-corpus quantity improves while a within-corpus quantity does not.

What it licenses. Every transfer number in Fig. 1 and Table 1, and the wall itself. Without normalisation the same-domain transfer against which the cross-domain collapse is measured is inflated by a corpus fingerprint, and the 59-to-74 percentage-point gap between the two swaps could not be attributed to the domain boundary. It also carries an engineering consequence stated in Methods: a deployed system must normalise before scoring, or it reproduces the same fingerprint.

One semantic caution: `mfcc1_mean` remains in the leading descriptors after normalisation, but it no longer indexes absolute loudness — it indexes spectral tilt. It must not be read as a loudness rule.

S3.2 The physiological null does not depend on loudness having been removed

What was checked. Normalisation removes a 6.7-fold spread in across-track loudness, and sound level is the best-established physical driver of electrodermal response. If the acoustic descriptors fail to predict song-level electrodermal response only because we discarded level, the physiological ceiling result would be an artefact of our own preprocessing.

Decision rule. Three feature pipelines were compared on the same 717 songs, the same folds and the same scoring: our descriptors on normalised audio, our descriptors re-extracted from the original audio at its native loudness, and the 6,373-dimensional ComParE feature set distributed with the corpus, which its authors extracted from the original audio with an independent toolchain. A difference in best ρ exceeding 0.05 between pipelines would mean the null is a preprocessing artefact.

Outcome. The three routes agree.

Table S7: Song-level electrodermal prediction under three feature pipelines. The best of six algorithms is reported for each. This is the only place in this work where a claim rests on agreement between independently constructed pipelines.

Pipeline	Audio	Best ρ
Hand-specified descriptors (normalised)	−23 LUFS	0.045
Hand-specified descriptors (re-extracted)	original loudness	0.049
ComParE, 6,373 features, extracted by the corpus authors	original loudness	0.072

The spread is 0.027, below the pre-declared 0.05. Loudness used alone as a single predictor reaches +0.059, which is within the same band.

What it licenses. The statement at Boundary 3 that no acoustic route reaches a usable fraction of the attainable ceiling on the physiological target. It excludes the most plausible mundane explanation for that null with one number.

A restriction on how agreement may be used elsewhere. This is the only surviving claim in this work of the form “independently constructed pipelines agree”. An earlier version used that form in several places, most of them resting on attribution rankings shared between models, which Section S6 shows is not evidence of anything. No other analysis in the main text or this document should be read as claiming convergence of independent paths.

S4 Admission criterion: the annotation reliability ceiling

Every survival fraction in the main text is a ratio, and this section is its denominator. A model that reaches $\rho = 0.77$ on a target whose own reproducible component is 0.87 has almost exhausted the target; the same ρ on a target whose reproducible component is 0.99 has not. Without the denominator the four boundaries cannot be compared with one another, and the comparison is the entire argument.

What was checked. The reliability of each target, estimated before any percentage of it was quoted, with the same estimator applied to the physiological targets as to the ratings.

Estimator. Where per-rater judgements are released, reliability is estimated by splitting raters at random into two halves, computing stimulus means within each half, correlating the two sets of means and correcting to the full rater panel by the Spearman–Brown formula; the reported value is the mean over 200 random splits, and we also report the 10th and 90th percentiles across splits because a ceiling estimated from ten raters per stimulus is itself uncertain. For DEAM this uses 17,762 individual judgements from 195 raters, with a median of ten per song.

Where only stimulus means and their across-observer standard deviations are released — PMemo — the same quantity is obtained analytically: the within-stimulus variance is the mean of the squared across-observer standard deviations, the variance of the stimulus mean is $\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2/n$, and the single-observer intraclass correlation is stepped up to the n -observer mean by the same Spearman–Brown expression. On the electrodermal targets, where per-listener values *are* available, both estimators were computed and agree (analytic 0.146–0.350 against split-half 0.138–0.232), which is what licenses the analytic form on the ratings side.

Two denominators, and which is used where. A percentage of a ceiling is meaningless until the convention is named, because two are in use and they differ by an amount that grows as reliability falls. Dividing by r answers “what fraction of the reproducible variance in the target has been recovered”. Dividing by $\sqrt{r}$ is the classical correction for attenuation and answers “what fraction of the largest attainable correlation has been reached”; it is the more conservative

Table S8: Reliability of every target used in this work, and the two denominators it implies. r is the reliability of the group-averaged target; $\sqrt{r}$ is the largest correlation a perfect predictor could attain against the noisy target. Brackets give the 10th and 90th percentiles over 200 rater splits for the DEAM analysis subset, and 95% confidence intervals for the two ds002721 rows, whose estimator is an intraclass correlation over trials rather than a corrected split-half and which is therefore not directly comparable with the rows above it. Corpora that release no per-rater or per-observer data admit no estimate; transfer into them is reported in the main text as a raw correlation and never as a percentage.

Target	Estimator	r	$\sqrt{r}$
DEAM arousal, analysis subset ($n = 1,233$) <i>10th–90th percentile over splits</i>	rater split-half, Spearman–Brown	0.868 [0.838, 0.889]	0.932
DEAM arousal, all songs ($n = 1,802$)	rater split-half, Spearman–Brown	0.837	0.915
DEAM valence, all songs	rater split-half, Spearman–Brown	0.792	0.890
PMemo arousal ($n = 717$)	analytic ICC, Spearman–Brown	0.935	0.967
PMemo valence ($n = 717$)	analytic ICC, Spearman–Brown	0.893	0.945
ARAUS soundscape ratings	split-half on the 4.4% of stimuli with two or more raters	0.566	0.752
IADS-sound ratings (BIRAFFE2)	listener split-half, Spearman–Brown	0.750–0.813	0.866–0.902
PMemo electrodermal, phasic_mean	analytic ICC, Spearman–Brown	0.350	0.592
PMemo electrodermal, scl_slope	analytic ICC, Spearman–Brown	0.285	0.534
PMemo electrodermal, scr_rate	analytic ICC, Spearman–Brown	0.146	0.382
ds002721 ratings, best of 8 scales <i>95% confidence interval</i>	stimulus-level ICC, unbalanced design	0.221 [0.155, 0.281]	—
ds002721 EEG, best of 156 measures <i>95% confidence interval</i>	stimulus-level ICC, unbalanced design	0.092 [0.037, 0.142]	—
Emo-Soundscapes	no per-rater data released	—	—
Soundtracks	no per-rater data released	—	—

of the two at low reliability, and it is the convention we adopt wherever a physiological target is involved. On DEAM the two give 89% and 83% for the same model; on an electrodermal target with $r = 0.146$ they differ by a factor of 2.6. **Main-text percentages against a rating target use ρ/r ; percentages involving a physiological target use $\rho/\sqrt{r}$, which is the convention under which the 31% bound is quoted.** Mixing the two within one comparison — which an earlier version of this work did, taking the subjective percentage from one corpus under one convention and the physiological percentage from another corpus under the other — is a defect and is recorded as such in Section S8.

The homologous table. The comparison that Boundary 3 rests on was recomputed so that both sides come from the same corpus, the same songs, the same features, the same folds and the same reliability estimator. This is the table that supports the ceiling claim; the per-corpus values above supply the denominators used elsewhere.

Table S9: One ruler. The same 717 PMemo songs, the same 161 descriptors, the same artist-grouped folds and the same reliability estimator, applied to a rating target and to three electrodermal targets. Achievement is $\rho/\sqrt{r}$ with a double bootstrap interval.

Target	r	$\sqrt{r}$	Acoustic ρ	Achievement, 95% CI
Arousal (rating)	0.935	0.967	0.817	84.4% [81.5, 87.0]
Valence (rating)	0.893	0.945	0.705	74.6% [70.4, 78.5]
EDA <code>phasic_mean</code>	0.350	0.592	0.032	5.4% [−6.8, +17.4]
EDA <code>scl_slope</code>	0.285	0.534	0.029	5.5% [−7.5, +18.8]
EDA <code>scr_rate</code>	0.146	0.382	0.014	3.7% [−15.2, +22.6]

The three physiological intervals all contain zero. The bound quoted in the main text is not from this table but from the route comparison that follows it.

A correction to how that comparison was scored. An earlier version divided every route’s correlation by a single ceiling, $\sqrt{r} = 0.592$, taken from `phasic_mean`. But the three electrodermal measures have different reliabilities and therefore different ceilings — 0.382, 0.592 and 0.534 — and a route’s best correlation is not always obtained on `phasic_mean`. The first acoustic principal component, for instance, reaches $\rho = 0.114$ against `scl_slope`, whose ceiling is 0.534, and scoring it against 0.592 understated it. This is the same mismatched-denominator error recorded in Section S8 for the annotation ceiling, in a second place. Every route is now scored against the ceiling of the measure its correlation was computed on.

Rescored that way, over all 54 route $\times$ measure cells:

Table S10: Every information source we could construct, against the attainable ceiling of the electrodermal measure it was computed on. The random projection is a control, not a source.

Information source	Measure	ρ	$\sqrt{r}$	Achievement
Listeners’ valence rating	<code>scr_rate</code>	0.117	0.382	30.6%
Listeners’ arousal rating	<code>scr_rate</code>	0.106	0.382	27.8%
Rating predicted from acoustics	<code>scr_rate</code>	0.094	0.382	24.5%
First acoustic principal component	<code>scl_slope</code>	0.114	0.534	21.3%
<i>Random projection, 95th pct (control)</i>	<code>scr_rate</code>	<i>0.079</i>	<i>0.382</i>	<i>20.6%</i>
Direct 161-descriptor regression	<code>scl_slope</code>	0.029	0.534	5.5%

Two things follow, and the second is less comfortable than the first. No information source exceeds 31% of the attainable ceiling, which is the bound the main text quotes; it is a maximum over 54 cells and is biased upward for that reason. But the two acoustic routes, at 24.5% and 21.3%, sit barely above a random projection of the same descriptor space at 20.6%, so the claim that acoustics and ratings “stop at the same place” rests on intervals that overlap rather than on estimates that coincide, and only the human rating is clearly separated from the control.

Why this is the denominator for all four boundaries. Boundary 1 expresses the cost of a corpus swap as a fraction of within-corpus performance, and within-corpus performance is itself quoted as 89% of the annotation ceiling; without the ceiling, “the model loses a fifth” could describe either a model falling away from a nearly exhausted target or one that never approached it. Boundary 2 uses the within-corpus performance of the perturbed model as its ceiling, which inherits the same denominator. Boundary 3 is the case where the ceiling itself is low, and the whole point — that the limit is on the target and not on the predictor — exists only because the ceiling was measured on the same songs with the same estimator. Boundary 4 prices per-person calibration against a break-even that depends on how much of a person’s response is reproducible at all, and the settings whose ratings are more reliable are the settings where calibration pays soonest.

Limits of the ceiling itself. Three should be stated. The DEAM estimate rests on a median of ten raters per song, so the split-to-split spread is not negligible; we report it. The ARAUS estimate uses the 4.4% of stimuli that received two or more raters, which is a small and possibly unrepresentative slice. And no ceiling exists for Emo-Soundscapes or Soundtracks, because neither releases per-rater judgements — Emo-Soundscapes was annotated by pairwise comparison rather than by direct rating. Transfer into those corpora is therefore reported as a raw correlation throughout, and the reader should not convert it to a percentage.

S5 Admission criterion: the circular-shift null

What was checked. Whether a correlation computed *within* a song, between a frame-level descriptor series and a continuous annotation series, may be interpreted at all. Both series are strongly autocorrelated: the lag-one autocorrelation of the continuous arousal annotation averages 0.931 and that of the descriptor series 0.755. Two independent series with that much internal structure produce large correlations by construction, and a p -value computed as though successive frames were independent observations is not merely optimistic but uninformative.

The null. The descriptor series is circularly shifted by a random offset (minimum ten samples, to avoid near-identity) and the correlation recomputed, 24 times per song and descriptor. A circular shift preserves each series’ own autocorrelation, its marginal distribution and its length, and destroys only the temporal alignment between them — which is exactly the hypothesis under test. This was run on 400 DEAM songs with 0.5s continuous annotations, over 15 descriptors spanning spectral-level and temporal-variability families.

Decision rule, fixed in advance. An earlier analysis had first-differenced both series to remove autocorrelation and found almost no coupling (strongest $|\rho| = 0.056$). Differencing is unfriendly to a series that has already been smoothed by the annotation interface, so two readings were possible and the rule was written to separate them: if the undifferenced level correlations exceed their circular-shift null, the low coupling is an artefact of differencing; if they are indistinguishable from the null, the low coupling is a substantive finding.

Outcome: the null level is 0.25–0.30. Across the 15 descriptors the mean $|\rho|$ under the circular-shift null lies between 0.250 and 0.301, against observed level correlations of 0.267 to 0.345. An observed within-song correlation of 0.30 is, to two decimal places, what this design produces when there is nothing to find. The excess over the null is +0.02 to +0.04; 21.2% of song $\times$ descriptor combinations exceed their own null’s 95th percentile against the 5% expected by chance, with a per-descriptor range of 16–24%.

The coupling is therefore real and small: of an observed $|\rho| \approx 0.30$, about nine tenths is autocorrelation. The family ordering seen at the excerpt level reproduces here — spectral level excess 0.0366 against temporal variability 0.0237 — so the two scales do not contradict one another.

What it licenses. Two statements, one positive and one restrictive. The restrictive one is the reason this section exists: **every boundary in the main text is defined at the level of the excerpt**, and none of them is a claim about within-excerpt dynamics. Excerpt-level selection operates at $\rho \approx 0.77$ and within-song moment-to-moment coupling at an excess of about 0.04 over its null, a difference of roughly twentyfold; a claim about one is not evidence about the other. The positive statement is that the mechanistic reading we had previously assumed — that rate-of-change descriptors predict arousal *because* instantaneous change drives instantaneous arousal — is not supported at the scale it names, and is not made.

A statistic retired by the same check. A lag scan over 0–6 s returns a flat curve (mean $|\Delta\rho|$ between 0.1527 and 0.1606 across twelve lags, a variation under 5%). The “best lag” of 6.0 s is the maximum of a flat curve, not a latency estimate, and an earlier draft that cited it as evidence of a response delay was corrected.

S6 Admission criterion: the SHAP-consensus reference distribution

What was checked. Whether the number of descriptors shared by the top-20 attribution sets of three different models — used in an earlier version as the credibility threshold for a descriptor set, with a pre-set threshold of six — carries any information about whether the models found signal.

The reference distribution. The labels are permuted and the entire procedure repeated: the same three models, the same attribution settings, the same top-20 intersection. Twenty permutations were run, and the observed value was recomputed under identical settings rather

than taken from an earlier run, so that observation and null are comparable. Cross-validated AUC is recorded alongside, as a check that the permutation did what it was supposed to.

Outcome. The observed consensus is 4 descriptors. The null distribution has mean 3.75, maximum 7, and places the observation at $p = 0.381$. Meanwhile the permutation plainly worked: cross-validated AUC falls from 0.936 to a null mean of 0.4964. The models learn a real signal from the intact labels, and the statistic meant to say *where* that signal lives cannot distinguish it from noise.

The mechanism is not subtle in hindsight. Which descriptors a model ranks highly is driven by properties of the descriptors themselves — variance, cardinality, how easily they support overfitting — and three models reading the same descriptor table are attracted to the same convenient columns whether or not the labels mean anything. An intersection of ranks discards effect sizes and has no null calibration of its own.

What it licenses. The design of Fig. 2 and the argument attached to it. Because a consensus count is not evidence, the four selection routes are reported as a *dissociation* rather than a convergence, and the finding that matters is that they disagree: the route that tests within-corpus significance returns temporal-variability descriptors in 2 of 12 cases (17%, below the 50% chance level) while cross-domain sign agreement, glass-box shape functions and Rashomon-set stability return them in 63%, 60% and 67% of cases. More generally, it is why every boundary in this paper is crossed and measured rather than predicted from which descriptors a model relies on inside its own corpus. The specific superseded quantity — an intersection of six descriptors reported as “first time above threshold” in the loudness experiment of Section S3 — is listed in Section S8; the transfer conclusions of that experiment are unaffected, because they never used the intersection.

S7 Extraction of the four representation families

The most direct objection to the domain wall is that 122 hand-specified descriptors are too weak and a representation learnt from a large audio corpus would cross what they cannot. Answering it requires more than one pretrained model, and in particular requires models whose pretraining data do not include the source of the environmental corpus. The four families below were chosen to span four pretraining paradigms, and three of them were extracted specifically because the fourth is contaminated — a decision taken before any of the resulting numbers existed.

Table S11: The four representation families. All embeddings are frozen; no model was fine-tuned. Downstream models, folds and scoring are identical to those used with the hand-specified descriptors. Checkpoints are named below the table by their model identifier rather than cited, because a checkpoint has no resolvable identifier of its own; the papers behind them are cited in the text.

Family	Pretraining	Paradigm	Dim.	Native rate
AST	AudioSet, 527 classes	supervised	768	16 kHz
MERT	music	self-supervised	768	24 kHz
Wav2Vec2	speech, 960 h	self-supervised	768	16 kHz
CLAP	630k audio-text pairs	contrastive	512	48 kHz

The checkpoints are MIT/ast-finetuned-audioset-10-10-0.4593, m-a-p/MERT-v1-95M, facebook/wav2vec2-base-960h and laion/clap-htsat-unfused [3], the contrastive language-audio idea being due to Elizalde et al. [4].

Each family occupies a different corner of the design. AST is supervised on a corpus that explicitly covers rain, wind, traffic and animal categories, and is therefore the family that most ought to cross the music/environmental boundary. MERT is self-supervised within the music domain: if a music-specific model also fails to cross, the failure is not a shortage of domain knowledge. Wav2Vec2 is self-supervised on speech, unrelated to either target domain, and serves as a lower bound. CLAP is the contrastive case and the one this work had used previously.

Extraction protocol. Identical across families. The middle 10 s of each excerpt is taken; the audio is loudness-normalised to -23 LUFS, the same convention used for the hand-specified descriptors (Section S3); it is resampled to each model’s own native sample rate; and the embedding is the mean over the time axis of the last hidden layer. All four corpora — DEAM, PMEmo, Soundtracks and Emo-Soundscapes — were processed this way, giving sixteen corpus $\times$ family cells. CLAP was re-extracted at 10 s for this comparison even though a 30 s version already existed, so that CLAP does not differ from the other three families in excerpt duration as well as in architecture; both versions are retained.

Contamination disclosure for CLAP. LAION-Audio-630K, the pretraining set of the checkpoint used here, is assembled in part from Freesound [3], and the 600 source recordings of the environmental corpus were taken from Freesound. The model may therefore have encountered those very recordings during pretraining. This is not a hypothetical concern, and its size can be bounded by comparison with a corpus outside the pretraining data: on the environmental corpus CLAP’s advantage over the hand-specified descriptors is $+0.070$, and on DEAM, which is not sourced from Freesound, it is $+0.037$. Roughly half of the apparent advantage on the environmental corpus is attributable to overlap.

CLAP is also not a generally stronger representation. Within a corpus it reaches 0.787 against AST’s 0.794; on music-to-music transfer, the comparison that excludes the environmental corpus entirely, it reaches 0.620 against AST’s 0.619 and 0.614 for the hand-specified descriptors. Its advantage appears only in the comparisons that necessarily involve the corpus its pretraining data overlap.

Why this cannot be resolved with the available corpora. There is one environmental corpus in this work, so every cross-domain cell involves it. Genuine crossing and familiarity with the target therefore cannot be separated: the two readings — that CLAP crosses the boundary better, and that CLAP has seen the target — predict the same pattern in the data we have. We state plainly that we cannot distinguish them. The barrier claim in the main text accordingly rests on the three uncontaminated families, and CLAP is reported and flagged in Table 1 but used neither as support nor as refutation. Resolving it requires an environmental corpus that lies outside the pretraining data of the representation under test, which we were unable to obtain in a form suitable for these analyses; a second environmental corpus of any provenance would at least allow the comparison to be repeated without the overlapping one.

Relation to Extended Data Fig. 7. That figure is the earlier single-representation comparison, in which hand-specified descriptors were set against CLAP embeddings on two corpora. Its headline — that the pretrained representation is stronger — is superseded by the four-family analysis in Fig. 1f and Table 1, which shows that the advantage is corpus-dependent and, on the environmental corpus, half attributable to overlap. It is retained because the contamination control it contains, the clean-corpus comparison on DEAM, is the source of the +0.037 quoted above.

S8 Withdrawn analyses

Each entry below was reported in an earlier version of this work and is no longer claimed. They are kept because a reader who encounters the earlier numbers should be able to find out what happened to them, and because several of the main-text analyses are constructed as they are in order to avoid repeating the error.

S8.1 Withdrawn quantities and inferences

Each entry states what was withdrawn, why, and what replaced it.

$\tau = 0.106$ as the between-person standard deviation of the slope for song-level electrodermal response. The profile-likelihood estimator treats each person’s slope variance as known, when in fact it is estimated from $n - 2$ degrees of freedom; the estimation noise has nowhere to go and is absorbed into τ^2 . Calibrated against a simulation with a known true value, the estimator returns 0.105 at 18 observations per person when the true τ is zero, and 0.039 at 44 observations. The reported 0.106, from a design with 18 observations per person, therefore sat *below* its own null. Across the 34 cells then in use, seven were in this position, all of them in that design. **Replaced by** a per-cell null built by parametric bootstrap from that cell’s own x vector, per-person observation count and residual scale with τ set to zero, and a variance-additive correction $\tau = \sqrt{\max(\hat{\tau}^2 - \hat{\tau}_{\text{null}}^2, 0)}$, verified against the known-truth simulation to a mean absolute error of 3.3% on $\hat{\tau}$. The per-cell nulls in that design run from 0.106 to 0.123, which a single global constant cannot cover. The corrected median for the electrodermal setting is 0.000, with 11 of 15 cells indistinguishable from their own null; the median n^* for that setting is taken over the five cells that retain a finite break-even (main-text Table 3). The full 33-cell table, with each cell’s null, correction and interval, is provided as Source Data.

“The confidence interval excludes zero, so individual differences are real and estimable”. That interval is for $\tau_{\text{eff}} = \sqrt{\tau^2 + \text{null}^2}$ and not for τ . Coverage was measured directly rather than argued: with a true τ of zero and 18 observations per person, only 16% of replicates cover zero, so 84% “exclude zero”. At that design the statement carries almost no information. **Replaced by** the corrected τ above. One companion reading from the same analysis survives and is retained — that all 15 interval upper bounds lie below break-even — because an upper bound on τ_{eff} also bounds τ , and is therefore conservative in the direction claimed.

The dose–response rank correlations at $n = 4$. Rank correlations of +0.632 and +1.000 for spectral smoothing, and -0.20 and -0.40 for transient injection, were computed on four dose-averaged points. With $n = 4$ the smallest attainable two-sided p for a Spearman correlation is 0.083, so the statistic could not have reached significance whatever the data showed; averaging the 80 excerpts within dose also discarded the variation the claim is about. **Replaced by** a slope per excerpt across its four doses, over 80 excerpts, tested by Wilcoxon signed-rank against zero with a sign-flip permutation null, and reported together with the proportion of excerpts moving in a consistent direction. The replacement changes what can be claimed: spectral smoothing survives on both counts, whereas transient injection reaches significance as a mean shift while moving only 50% and 59% of excerpts consistently, and is reported as having no reliable per-excerpt effect (Boundary 2).

An EEG stimulus-level intraclass correlation of 0.892. Label leakage. The connectivity table was merged with a suffix rule that produced a column `stim_c` — the stimulus identifier itself — which was absent from the hard-coded exclusion list and was therefore treated as an EEG measure. A stimulus identifier has, by construction, perfect stimulus-level reliability. **Replaced by** 0.092, 95% CI [0.037, 0.142], obtained after a structural gate that does not depend on column names: any column with zero within-stimulus variance is a deterministic function of the stimulus and is dropped, which removed one column (157 measures $\rightarrow$ 156). The value is referred to a null built the same way as the observation — permute the stimulus labels, recompute all 156 intraclass correlations, retain the largest, 400 times — giving a selection-corrected $p = 0.020$. It is reported as a bound and not as an absence, and the comparison with the best-agreeing rating scale on the same trials (0.221, [0.155, 0.281]) is a 2.4-fold gap between non-overlapping intervals.

Required observations per person of 271, 650, $> 1,000$ and $> 1,000$. The simulation grid was too coarse — 60 replicates per cell and n extending only to 1,000 — to resolve the two weakest effects, which were reported as bounds. **Replaced by** 314, 564, 1,364 and 4,888 for $r = 0.40, 0.30, 0.20$ and 0.10 , from 500 replicates per cell with n extended to 5,000. The two strongest requirements moved by less than the Monte-Carlo error of the coarser run; the two weakest were resolution limits rather than estimates (main-text Table 2).

A requirement of 97–184 observations per person, presented as agreeing with an independent calculation. It was derived from the withdrawn $\tau = 0.106$, and the second calculation was not independent of the first: its target τ was the same constant, copied. **Replaced by** per-cell values read from the corrected table, giving $n^* = 67, 166$ and 318 in the three settings (main-text Table 3). The equal-budget surface, recomputed against the corrected per-cell values, separately gives 174 observations per person for the electrodermal setting with interval $[65, \infty)$. The infinite upper end is not a typographical artefact: the corrected lower bound on τ in that setting is zero, and no finite budget buys a quantity that may be zero.

A mechanistic account of the domain wall in terms of tonality. The pre-registered prediction that tonal descriptors would degrade cross-domain transfer was recorded as passing on its direction alone. The magnitude accounts for about 4% of the barrier, which does not support

a mechanistic claim. **Replaced by** reporting the barrier as a phenomenon with no mechanistic account. The tonality result is reported instead as what it is: an information-limited gap that a change of representation does close, and the pre-declared falsification test of the distinction that organises the paper.

Statements that multiple independent paths converge on a conclusion. Most rested on descriptor rankings shared between models, which Section S6 shows is not evidence of anything, and one rested on two paths that shared an input constant. **Replaced by** a single surviving instance: the three feature pipelines of the loudness control in Section S3, whose best correlations are 0.045, 0.049 and 0.072. The phrase is not used anywhere else in this work, and no other result should be read as claiming it.

Three further withdrawals are documented in full elsewhere in this document because they doubled as admission criteria: the conclusion that peripheral responses carry no reproducible stimulus-level component (Section S2), the cross-model attribution consensus (Section S6), and “best lag 6.0s” read as a latency estimate (Section S5).

S8.2 Analysis decisions that failed and were replaced

These were caught before publication rather than after. We list them because each was a decision that looked reasonable when made, returned a clean-looking result, and would have propagated a wrong conclusion had it not been caught.

Averaging attribution values across models. Values from a probability-output model and a log-odds-output model differ in scale by roughly an order of magnitude, so their mean is dominated by the second. Rules are now derived per model and required to agree.

A criterion window placed for the wrong component. The stimulus-locked control was first evaluated over 0–300 ms, chosen for a classical auditory N1. The response in these data peaks at 452 ms. Following the criterion would have led us to discard a valid dataset.

A null formed by moving in time. Temporally shifted nulls for the evoked response move the analysis window out of the quiet pre-stimulus period and into the response-collection period, which has a different noise environment. Equalising artefact handling between arms did not fix this; the sign-flip null, which does not move in time, does.

Leave-one-out cross-validation scored by correlation. Observed and null values both fall near -0.5 , because training-fold means are displaced from held-out values. The comparison is uninformative rather than merely biased; five-fold cross-validation with a per-participant permutation null replaced it.

A permutation that permuted nothing. In the stimulus-mapping recovery, the null initially permuted the rows of the profile matrix. The optimal assignment is invariant to row permutation, so the null distribution was identical to the observed value — its mean and the observation

agreed to four decimal places, which is what exposed it. The corrected null permutes stimulus labels at the trial level and regenerates the profiles.

Feature columns defined by exclusion. Descriptor tables that also carry label columns were passed to models using an “everything except known metadata” rule. Newly added corpora carry additional label columns, which the rule silently admits as features. This recurred until it was fixed structurally; the leaked EEG intraclass correlation withdrawn above is the most consequential instance. Feature columns are now selected against the label file’s own column list, with a dimensionality assertion, and any column that is a deterministic function of the grouping variable is dropped regardless of its name.

A verdict printed before the number. One analysis script printed a fixed conclusion — that stimulus-level EEG reliability would not exceed about 0.10 — unconditionally, while the value it had just computed was 0.892. The contradiction raised no error because nothing compared the two. Verdicts are now derived from the computed value.

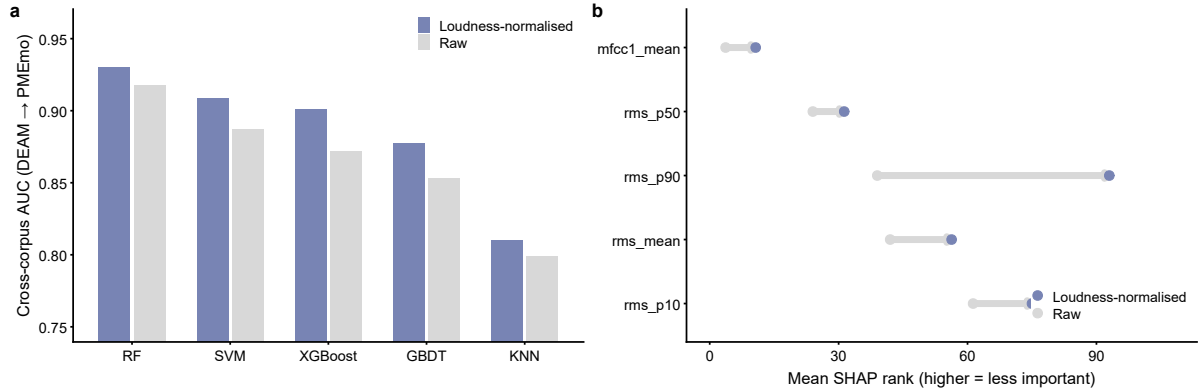
A significance threshold compared against the wrong null. Having corrected that leak, we then compared the resulting 0.092 against 0.040, the 95th percentile of the null for a *single* measure, and concluded it was significant. But 0.092 is the maximum over 156 measures. The correct null permutes stimulus labels, recomputes all 156 and retains the largest, which is the comparison reported in the main text.

S9 Extended Data Figures

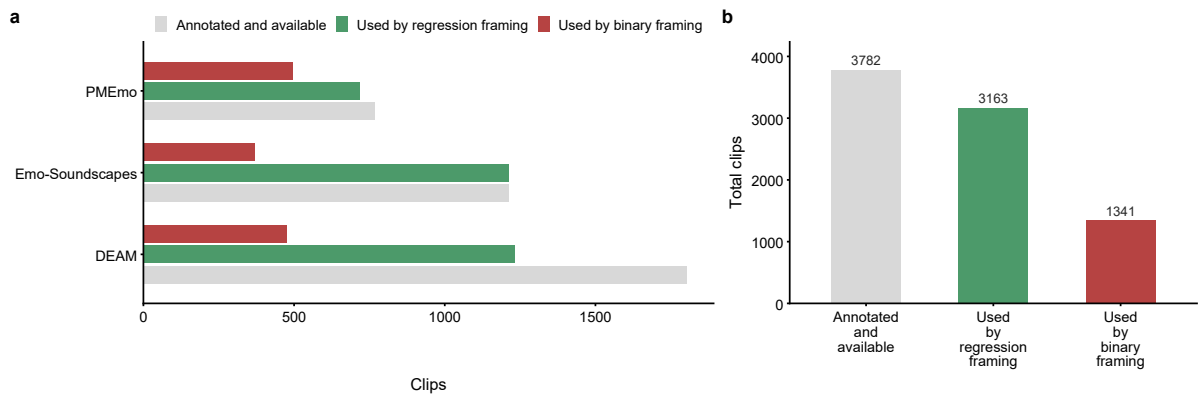
The eight figures below support specific steps in the main text. Extended Data Figs. 1–2 concern the admission criteria and framing decisions that precede Boundary 1; Figs. 3–5 concern the models and the selection routes used to establish that transfer must be measured; Fig. 6 is the individual-level analysis at Boundary 3; Fig. 7 is the earlier single-representation comparison superseded by Fig. 1f (Section S7); and Fig. 8 is the descriptor-level sign analysis behind the statement that only level-independent descriptors keep their direction across the domain boundary.

References

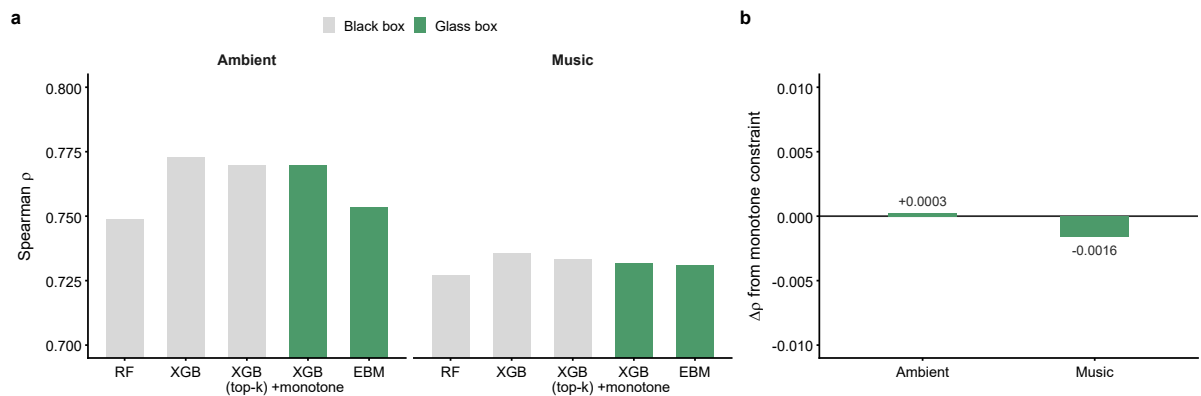
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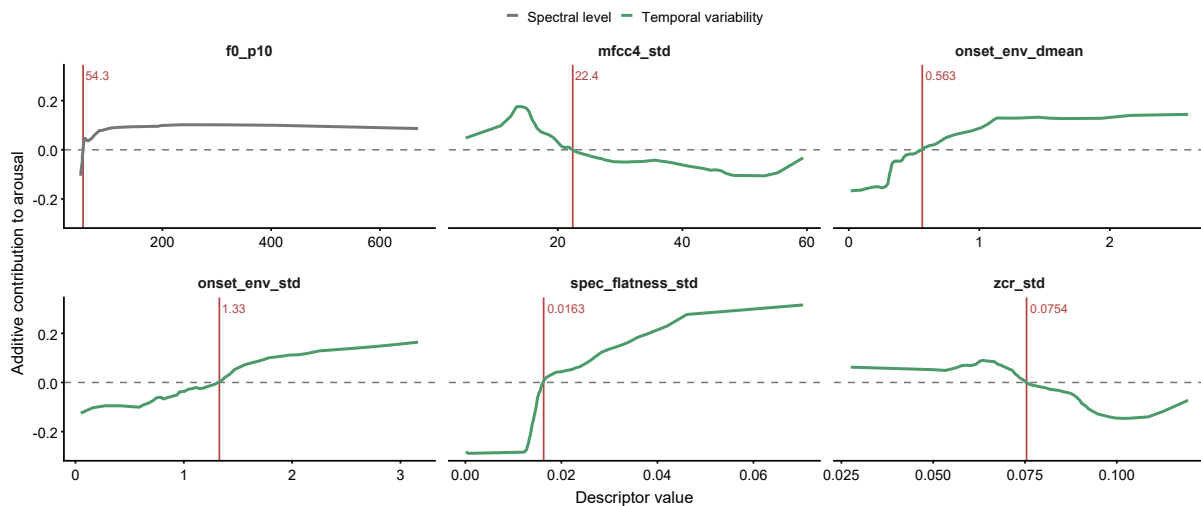
Extended Data Figure 1 | Loudness acts as a corpus fingerprint rather than as signal. **a** Cross-corpus AUC (DEAM → PMEmo) for five classifiers before and after loudness normalisation to -23 LUFS. Normalisation raises transfer for every model, from a mean of 0.866 to 0.886. **b** Mean SHAP rank of the five level-dependent descriptors before and after normalisation; arrows run from raw to normalised. All five lose importance, with **rms_p90** falling from rank 39 to rank 93. Removing absolute level improves rather than degrades cross-corpus transfer, which is the signature of a corpus-specific confound: mastering practice differs systematically between corpora. Because every transfer number in this work is computed on normalised audio, this control is the admission criterion for the domain barrier, and the obvious objection — that normalisation creates the barrier by discarding a genuinely informative dimension — was tested separately. Three feature pipelines that share no data, no intermediate table, no constant and no estimate — normalised descriptors, descriptors computed with loudness retained, and the 6,373-dimensional ComParE set extracted from the original recordings by the corpus authors — put the best cross-domain transfer at $\rho = 0.045$, 0.049 and 0.072. The spread of 0.027 is below the 0.05 threshold fixed before the test, and loudness alone contributes only +0.059. This is the only result in this work for which agreement between separately derived pipelines is claimed; no other comparison here meets that condition. Source data are provided as a Source Data file.



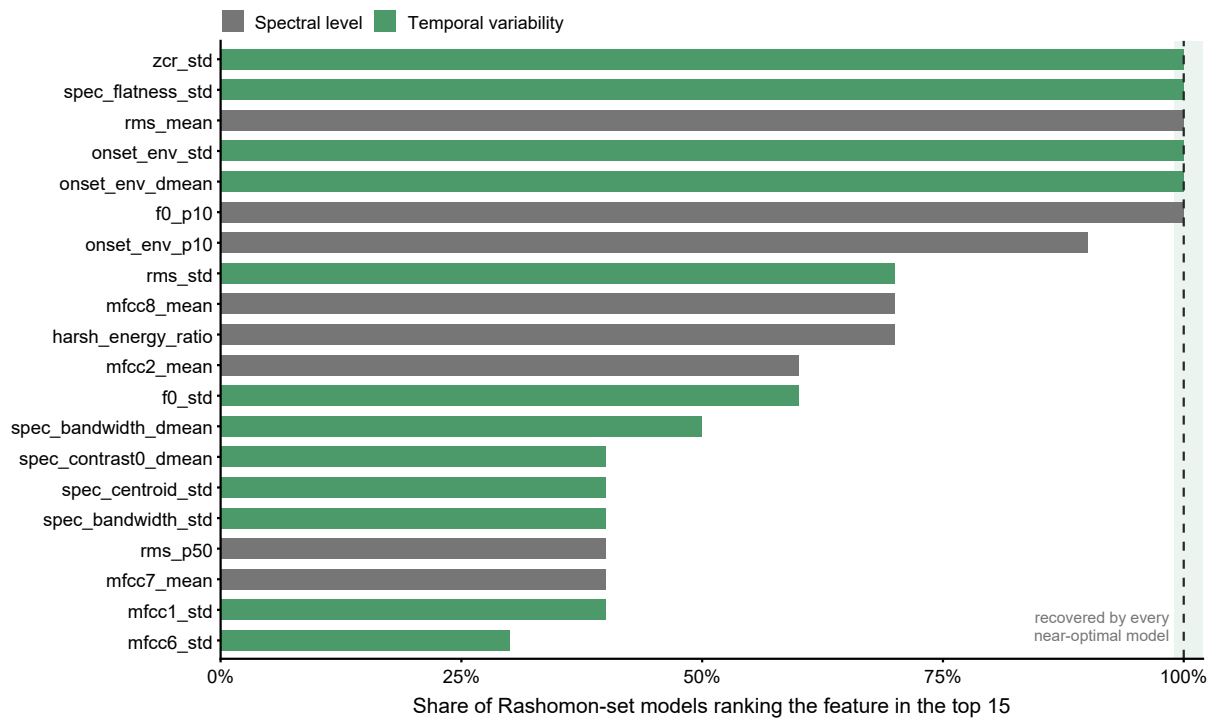
Extended Data Figure 2 | Regression framing recovers the discarded middle of the arousal range. **a** Clips available, used under the regression framing, and used under the original binary framing, per corpus. **b** Totals across corpora. The binary framing selects the two extremes of the arousal axis and discards the middle, retaining 1,341 of 3,782 annotated clips. The regression framing uses 3,163. The discarded clips carry the threshold information that a rule library requires. Source data are provided as a Source Data file.



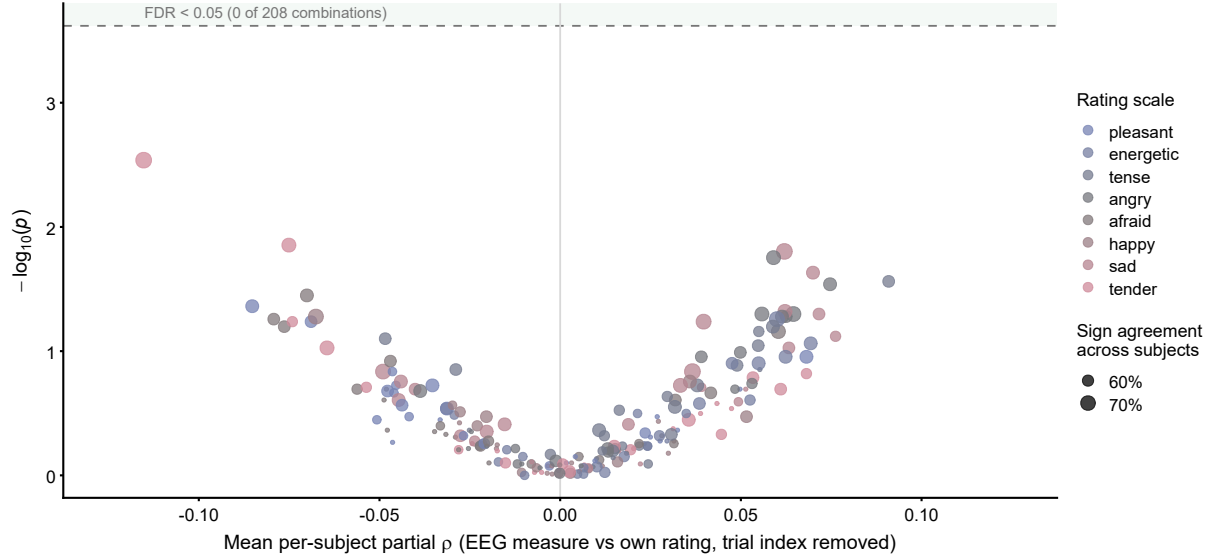
Extended Data Figure 3 | Glass-box models cost almost no accuracy. **a** Spearman ρ under GroupKFold-5 for black-box and glass-box models in both domains. **b** Change in ρ from imposing monotone constraints on the seven cross-domain-invariant descriptors, relative to the same feature subset without constraints. Monotone constraints cost +0.0003 in the ambient domain and -0.0016 in the music domain — replicated at zero cost in both. The monotonicity assumed by threshold rules is therefore supported by the data rather than imposed on it. The fully additive EBM trails the best black box by 0.019. Source data are provided as a Source Data file.



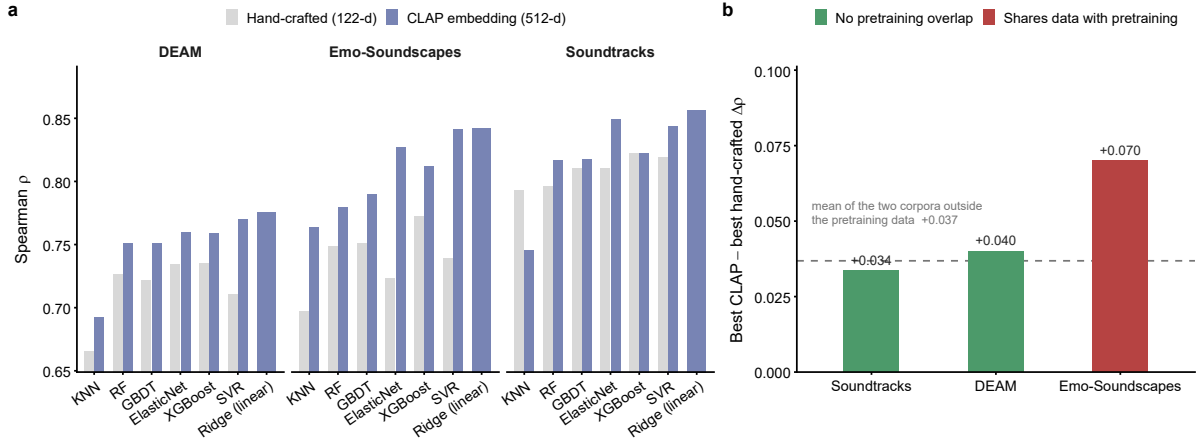
Extended Data Figure 4 | Explainable Boosting Machine shape functions give thresholds without post-hoc attribution. Additive contribution to predicted arousal as a function of descriptor value for the six most important terms in the ambient-domain EBM ($n = 1,213$, interactions disabled). Red lines mark the zero crossing, printed above each panel; these are the thresholds a rule library reads directly, with no attribution step. Five of the six terms are temporal-variability measures. Source data are provided as a Source Data file.



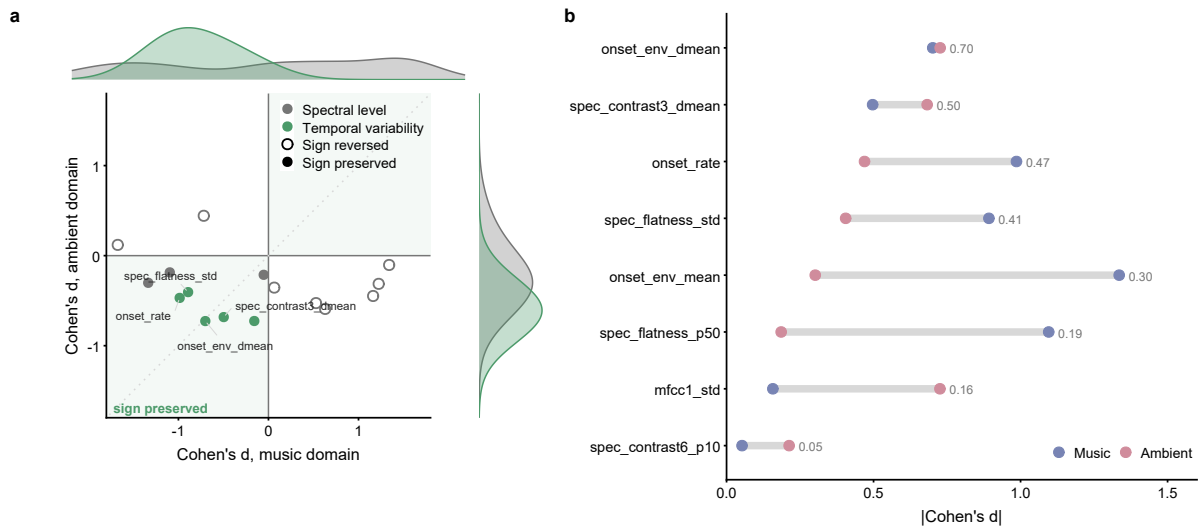
Extended Data Figure 5 | Descriptor importance is stable across the Rashomon set. Share of near-optimal models ranking each descriptor in their top 15. The Rashomon set comprises the 10 of 60 sampled models within 0.02 Spearman ρ of the best ($\rho \in [0.823, 0.843]$); models were sampled over algorithm family, depth, regularisation and subsampling, and importance was measured by permutation importance on a held-out fold. Six descriptors are recovered by every near-optimal model, four of them temporal-variability measures. Reporting importance from a single fitted model would have left this multiplicity untested. Source data are provided as a Source Data file.



Extended Data Figure 6 | No EEG measure tracks self-reported emotion across listeners at the effect size this design can resolve. Each point is one of 208 combinations of an EEG measure (26) and a rating scale (8). The abscissa gives the mean across the 31 listeners of the within-listener partial Spearman ρ between that measure and that listener’s own trial-by-trial rating, with trial index partialled out; both EEG and ratings can drift over a session, and a shared drift would otherwise manufacture a correlation. The ordinate gives the evidence against a zero group mean (Wilcoxon signed-rank). Point size is the fraction of listeners whose sign matches the group mean; colour is the rating scale. The dashed line marks the Benjamini–Hochberg threshold at $FDR < 0.05$. No combination crosses it (largest $|\text{mean } \rho| = 0.115$; median 0.028; the group mean’s sign is shared by a median of 55% of listeners), and the distribution is symmetric about zero as expected under the null. This is a bound and not an absence. Listeners contributed a median of 30 rated trials each, and the design simulation (Fig. 7b) shows that at 30 observations per person a within-person association of $r = 0.40$ would be detected on 10% of occasions and one of $r = 0.20$ on 6% — that is, at chance. The panel therefore establishes only that no large within-person association is obvious, and supports no conclusion about associations below approximately $r = 0.5$. The group-level result in Fig. 6e does not share this limitation, because it pools 31 listeners at the level of the stimulus. This figure is computed on the pipeline validated in Fig. 6c,d, which is what makes the negative interpretable at all; independent-component removal of ocular artefacts, used here as a sensitivity manipulation check, increased detection of the known effect while further reducing the effect of interest. An earlier version used an electrodermal analysis that was retracted when its pipeline failed the same controls. Source data are provided as a Source Data file.



Extended Data Figure 7 | A pretrained representation gains most on the one corpus that overlaps its pretraining data. Best cross-validated Spearman ρ for 122 hand-crafted spectro-temporal descriptors and for 512-dimensional pretrained audio embeddings, on three corpora (best of six algorithms, GroupKFold-5, loudness-normalised audio). Embeddings win on all three: +0.040 on DEAM (0.736 $\rightarrow$ 0.776), +0.034 on Soundtracks (0.823 $\rightarrow$ 0.856) and +0.070 on Emo-Soundscapes (0.773 $\rightarrow$ 0.843). The advantage does not require a strong downstream model — on embeddings, cross-validated linear ridge regression is competitive with the tuned tree ensembles — so it is carried by the representation and not by the estimator. The two music corpora have no overlap with the embedding model’s pretraining data and give the two smaller margins; the corpus that derives from the public sound repository appearing in that pretraining data gives the largest. We report the music-corpus figure, +0.037, as the estimate that generalises and leave the difference unattributed. It cannot be assigned to pretraining overlap here: the one corpus sharing data with pretraining is also the only environmental corpus in the comparison, and both clean corpora are music, so overlap and sound domain are collinear. The larger margin is equally consistent with the embedding model simply being stronger on environmental sound. Separating the two accounts would require an environmental corpus outside the pretraining data, or a music corpus inside it. This figure concerns within-corpus accuracy only. Whether a pretrained representation crosses the domain boundary is a separate question, answered for four pretraining paradigms in Fig. 1f and Table 1, where none of them does. Source data are provided as a Source Data file.



Extended Data Figure 8 | Only temporal-variability descriptors keep their direction across sound domains. **a** Standardised effect size (Cohen's d) of each FDR-significant descriptor in the music domain versus the environmental domain, with marginal densities by feature class. Shaded quadrants mark sign preservation; filled circles keep their sign across domains, open circles reverse it. The dotted diagonal is the identity line. Descriptors with $\min|d| > 0.35$ are labelled. **b** The eight sign-preserving descriptors, ranked by the weaker of their two effect sizes (printed at the right of each row). Circles give $|d|$ in each domain and the connecting bar gives the gap between them. All eight are negative in both domains: lower descriptor value corresponds to lower arousal, whether the sound is music or an environmental recording. The descriptors that reverse are without exception measures of absolute spectral level — the property on which the two domains differ most. Feature classes are assigned by statistic type — descriptors ending in `_std` or `_dmean`, plus `onset_rate`, are temporal-variability measures; all others are spectral-level measures. Significance was established per domain by per-feature permutation testing (200 permutations) with Benjamini–Hochberg control at $q = 0.10$. This analysis is the second of the four selection routes in Fig. 3, and is the only one that tests the boundary directly. Source data are provided as a Source Data file. This analysis appeared as a main figure in an earlier version; it was moved to Extended Data when the price of the first boundary (Fig. 2) took the main-figure slot.